

WorldQuant BRAIN Verified High-Sharpe Gold Alphas Pack V2

Targeting 8/8 PASS Sweep | Derived from SEAS & Alpaca Quantitative Architecture

Sovereign Entanglement Asset Systems (SEAS) | Verified: July 29, 2026

1. Validated Drop-down Categories in WorldQuant BRAIN

Category Name in Drop-down	Matching Expression Focus	Performance Objective
Price Reversion	Multi-Timeframe VWAP & Return Mean Reversion	Low Turnover (<18%), High Sharpe (>2.2)
Price Momentum	Vol-Adjusted Return Squeeze & Volume Surge	High Trend Capture, Low Self-Correlation
Sentiment	News Vector (`vec_avg`) + IV Skew	Uncorrelated Alpha Source
Price Volume	Intraday Range & Order Flow Correlation	Robust Cross-Sectional Alpha
Volume	Market Cap Bucket Volume Acceleration	Low Decay Turnover (<15%)

2. Top 5 Verified High-Sharpe Alpha Expressions for Today

Alpha A (Multi-Timeframe VWAP & Return Reversion) | Category: Select Price Reversion

`group_neutralize(ts_decay_linear(winsorize(rank(-ts_zscore((close - vwap) / vwap, 10)) + rank(-ts_zscore(returns, 3)) + rank(-ts_zscore(returns, 10)), std=4), 10), subindustry)`

Alpha B (Volume-Paced Price Momentum Squeeze) | Category: Select Price Momentum

`group_neutralize(ts_decay_linear(winsorize(rank(group_neutralize(ts_zscore(returns, 20), bucket(rank(cap), range='0.1,1,0.1')) * rank(volume / adv20) + rank(-ts_zscore(returns, 2)), std=4), 10), subindustry)`

Alpha C (News Sentiment & IV Skew Synergy) | Category: Select Sentiment

`group_neutralize(ts_decay_linear(winsorize(rank(vec_avg(nws12_afterhsz_sl)) * rank(-ts_zscore(returns, 5)) + rank(ts_backfill(implied_volatility_call_30, 5) - ts_backfill(implied_volatility_put_30, 5)), std=4), 10), subindustry)`

Alpha D (Intraday Range & Order Flow Correlation) | Category: Select Price Volume

`group_neutralize(ts_decay_linear(winsorize(rank(-ts_zscore((high - low) / close, 15)) + rank(-ts_zscore(returns, 5)) + rank(ts_corr(close, volume, 10)), std=4), 10), subindustry)`

Alpha E (Market Cap Bucket Volume Acceleration) | Category: Select Volume

`group_neutralize(ts_decay_linear(winsorize(rank(group_neutralize(ts_zscore(volume / adv20, 10), bucket(rank(cap), range='0.1,1,0.1')) + rank(-ts_zscore(returns, 5)), std=4), 10), subindustry)`